Surface developments, true shapes & intersections

GRAPHICS WITH MR KELLY

Higher · Drawing & spatial awareness · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify correct surface developments of prisms, cylinders, pyramids and cones
- ☐ interpret common geometric forms with single cuts
- ☐ describe interpenetration and intersections of prisms and cylinders

I DO — Worked example

Read how a full-mark answer is built

A cylindrical sweet tube is 40 mm in diameter.

1 Calculate the length of the rectangle in its surface development, to the nearest mm. (1 mark)

Points that earn marks from the marking instructions: one ✓ per point

✓ π × 40 = 126 mm (125.7)

WE DO — Guided practice

Use the starters and word bank to help you

A company prints thousands of card packaging nets for a phone.

2 Describe two ways the production of the packaging could be made more environmentally friendly. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

possible · reduce · environmentally · friendly · vegetable · recycled · sustainable · thickness

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A cone develops into…

☐ A long rectangle ☐ A complete circle ☐ An equilateral triangle ☐ A sector of a circle

b) Fitting as many nets as possible onto a sheet…

☐ Reduces waste and cost ☐ Increases waste ☐ Makes folding harder ☐ Is not allowed

c) A cylinder cut at an angle has a true shape that is…

☐ A circle ☐ An ellipse ☐ A rectangle ☐ A parabola

d) The length of a cylinder's development is…
☐ The diameter ☐ The radius ☐ π × diameter ☐ 2 × height

4 Key terms: write the letter of the matching description next to each term.

1. **Surface development**	____	**A** Extra flap on a net to join edges.
2. **Cylinder development**	____	**B** Rectangle of length π × d, plus two circles.
3. **Cone development**	____	**C** The flat net that folds to make a 3D form.
4. **Glue tab**	____	**D** A sector with radius equal to the slant height.

Two cylindrical pipes of different diameters meet at 90°.

5 Describe how the curve of intersection is found on the elevation. (3 marks)

6 Explain what is meant by 'Line of intersection'. (2 marks)

7 Explain what is meant by 'True shape of an angled cut through a cylinder'. (2 marks)

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Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?